

F. Test Plan and Automation

Application: SauceDemo (<https://www.saucedemo.com/>)

1. Objective

Create and execute a structured QA strategy for SauceDemo by:

- Defining scope, risks, and test approach
- Designing functional/regression/performance-oriented test coverage
- Automating key flows using Playwright
- Reporting actionable pass/fail outcomes and defects

2. Scope of Testing

In Scope:

- Authentication (valid, invalid, locked-out users)
- Inventory page (product visibility, sorting, add/remove cart actions)
- Cart and checkout (cart update, checkout info validation, order completion)
- Navigation (menu, logout, back/cancel paths)
- Basic UI/UX validation (labels, error messages, button states)
- Cross-browser smoke (Chromium baseline; optional Firefox/WebKit)

Out of Scope:

- Backend database validation
- Security penetration testing
- Deep load/stress testing at infrastructure level
- Mobile native app validation

3. Test Objectives

- Confirm core purchase journey works end-to-end for valid users
- Ensure invalid and edge-case inputs return correct behavior/messages
- Prevent regressions in high-risk user flows
- Build a maintainable automation baseline for repeatable execution

4. Types of Testing

- Functional Testing
- Regression Testing
- Negative Testing
- UI Validation
- Smoke Testing
- Basic Performance Observation (page load responsiveness, not full perf benchmarking)

5. Test Approach

- - Risk-based prioritization:

- High: Login, add to cart, checkout completion
- Medium: Sorting behavior, cart quantity changes
- Low: Non-critical UI text and cosmetic checks
- Manual review for exploratory findings
- Automated coverage for stable high-value workflows

6. Test Environment

- **Test URL:** <https://www.saucedemo.com/>
- **Test Data Users:**
 - standard_user
 - locked_out_user
 - problem_user (optional robustness checks)
- **- Browsers:**
 - - Primary: Chromium
 - - Secondary: Firefox/WebKit (optional smoke)
- **- Tools:**
 - Playwright
 - Node.js + npm
 - Git for version control
 - Optional: Allure/HTML report viewer

7. Resources and Roles

- **QA Engineer:**
 - Test planning, test case creation, execution, defect reporting
- **Automation QA:**
 - Framework setup, test scripting, CI readiness, reporting
- **Developer:**
 - Defect fixes, root-cause analysis support
- **Product/Stakeholder:**
 - Requirement clarification, UAT alignment

8. Entry and Exit Criteria

Entry Criteria:

- Application accessible in test environment
- Test credentials available
- Core requirements defined

Exit Criteria:

- 100% execution of planned critical test cases
- No open blocker/critical defects for release
- Regression suite pass rate $\geq 95\%$ (or accepted deviations documented)

9. Risks and Mitigation

- **Risk: Flaky tests from timing/AJAX updates**
 - - Mitigation: Use Playwright auto-wait + explicit state assertions
- **Risk: Test data instability**
 - - Mitigation: Reset app state before each test
- **Risk: Scope creep**
 - - Mitigation: Freeze sprint scope and track additions separately

10. Timeline and Milestones (Sample: 5 Working Days)

Day 1:

- Finalize test scope and test plan
- Define test data and acceptance criteria

Day 2:

- Create detailed manual test cases
- Stakeholder review/sign-off

Day 3:

- Set up Playwright framework
- Automate login and inventory flows

Day 4:

- Automate cart and checkout flows
- Add reusable utilities and reporting

Day 5:

- Full execution (manual + automation)
- Defect logging and final summary report

11. Deliverables

- Approved Test Plan
- Test Case Document
- Playwright automation suite
- Execution report with pass/fail summary
- Failure screenshots/logs and issue register